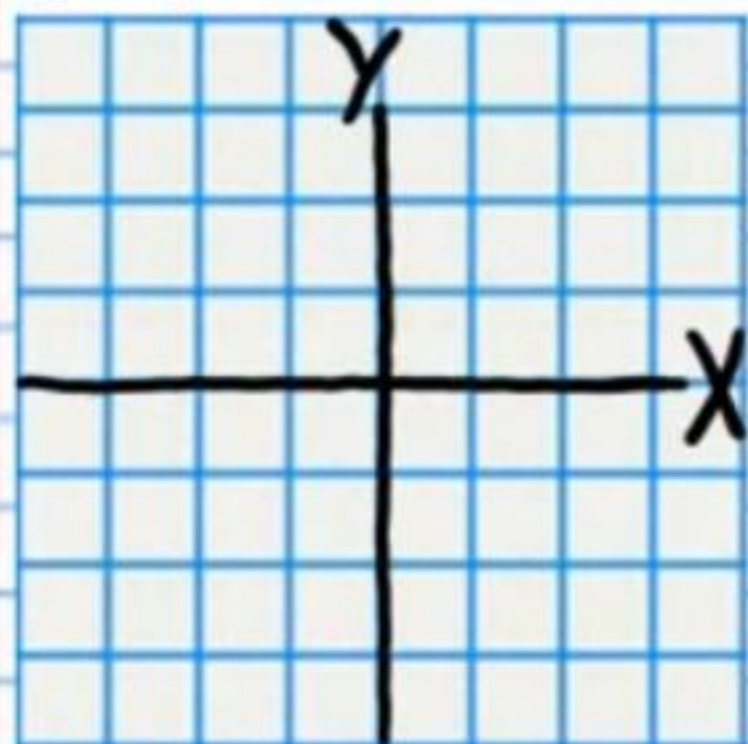


Polar Coordinates (Homework)

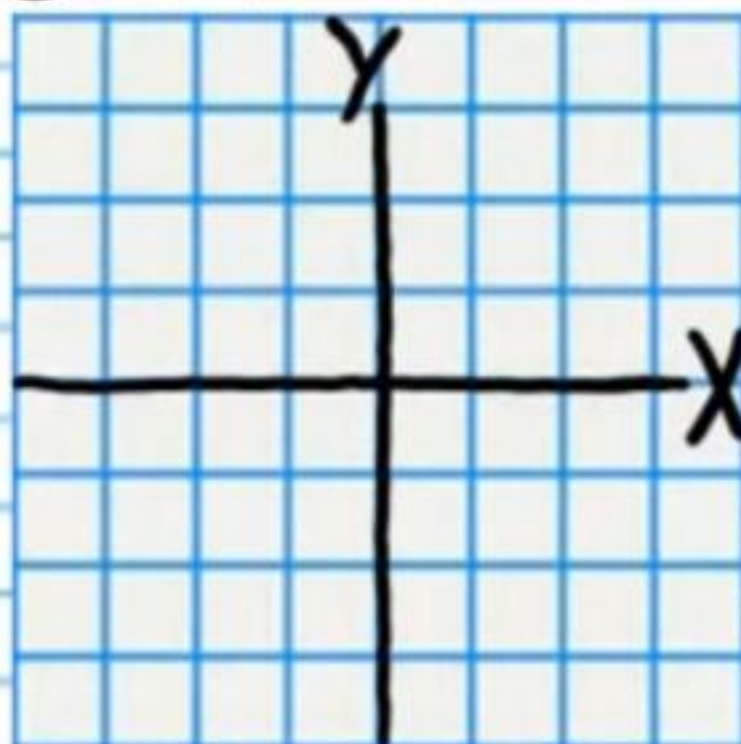
Name
Date
Course

Draw the points described by the polar coordinates below.

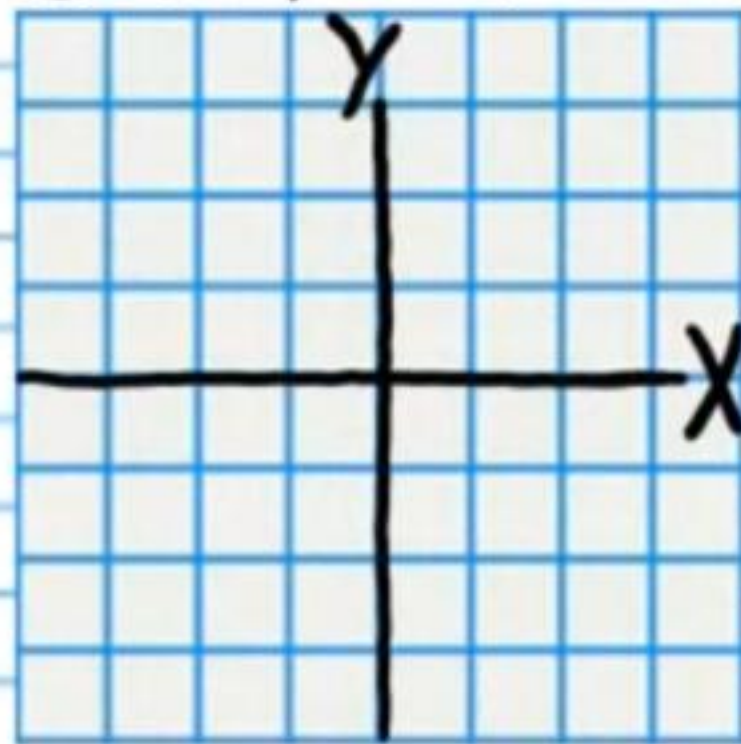
• ① $(1, 50^\circ)$



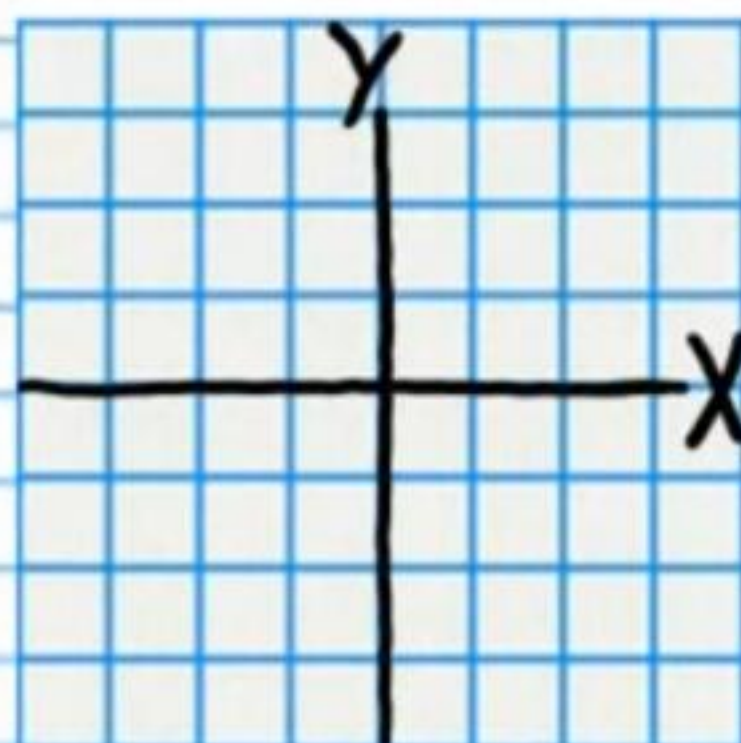
② $(2, -70^\circ)$



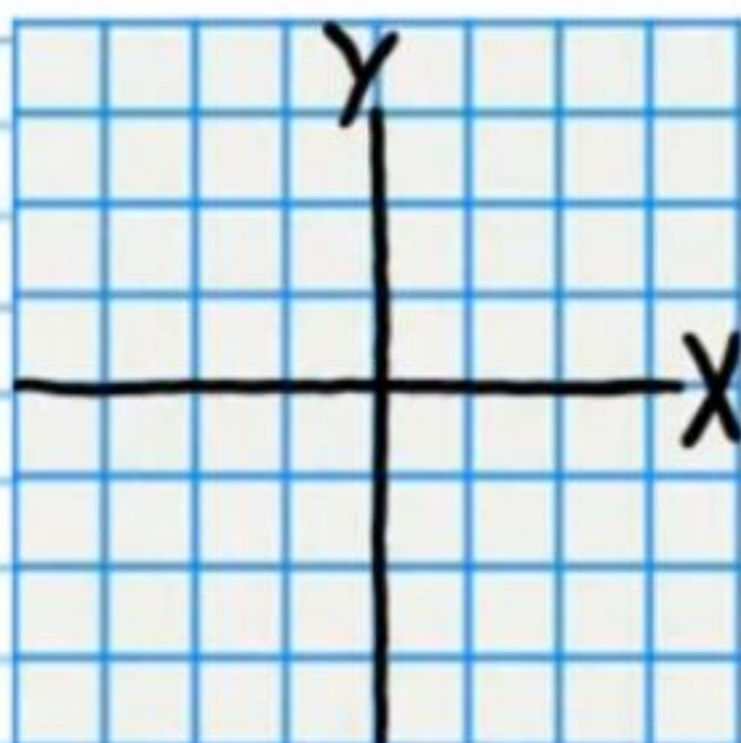
③ $(-3, 300^\circ)$



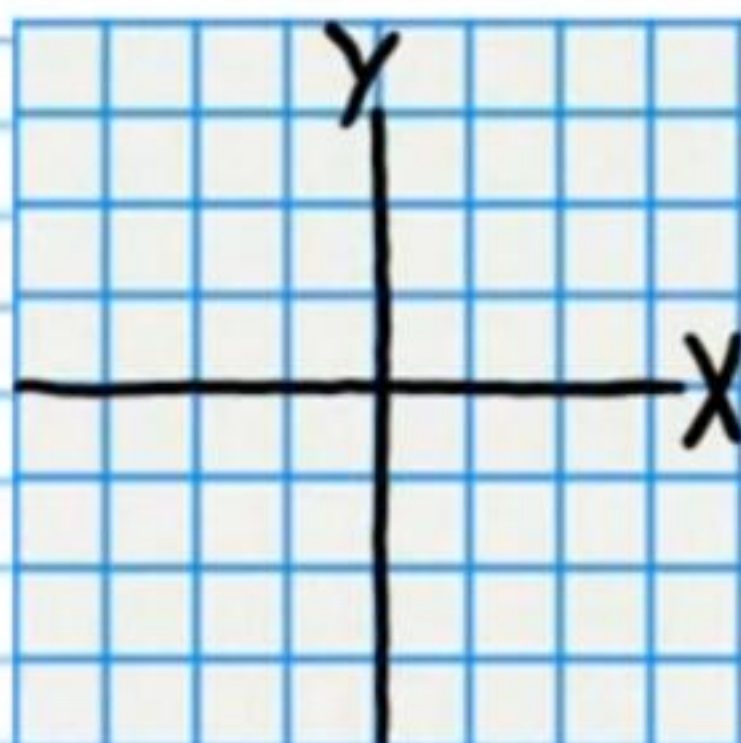
④ $(-2, -160^\circ)$



⑤ $(2.5, 235^\circ)$



⑥ $(-1, 260^\circ)$



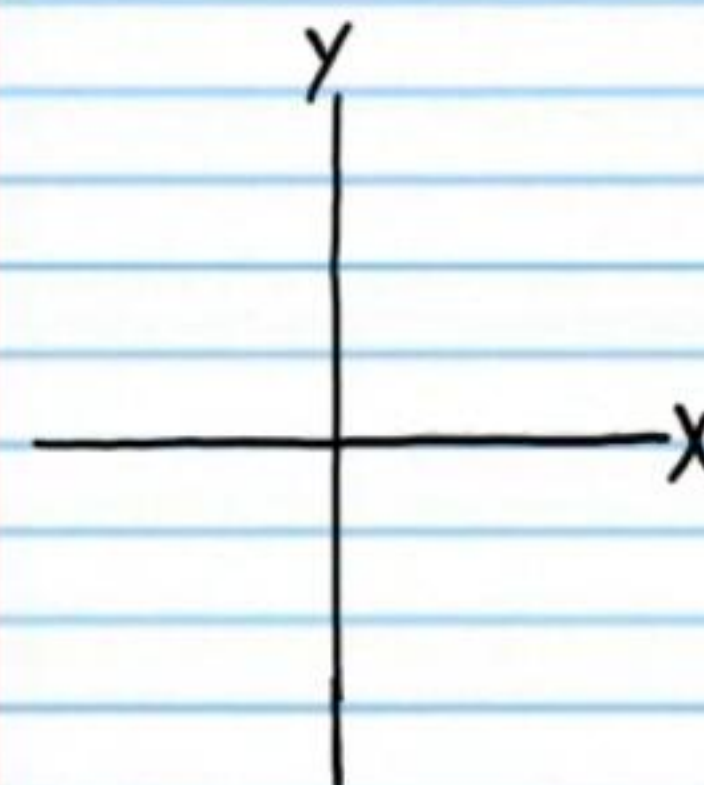
Convert the rectangular coordinates below to polar coordinates.
Use degree units for angle measures. Round to the nearest tenth if necessary.

• ⑦ $(4, -7)$

Find r .

Find θ .

Graph

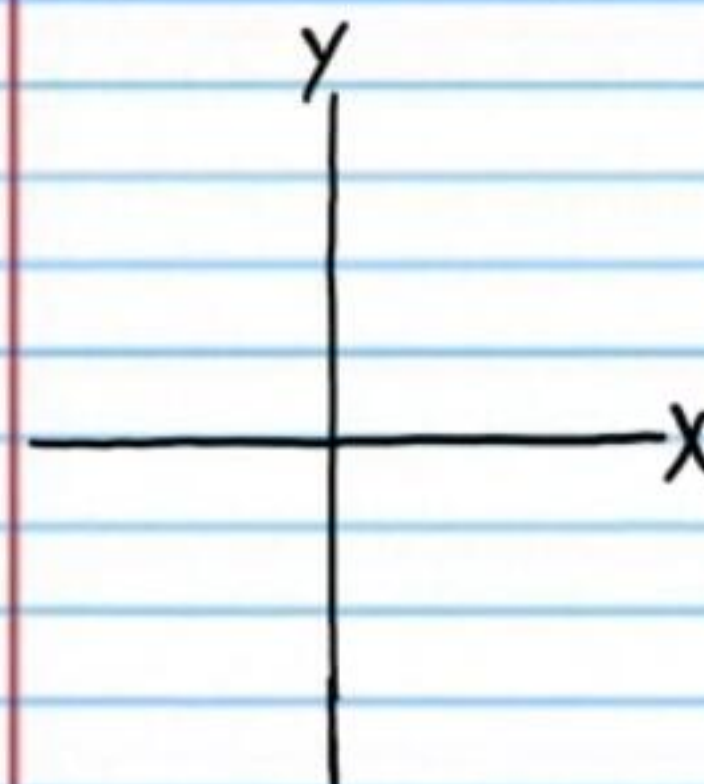


⑧ $(6, 5)$

Find r .

Find θ .

Graph

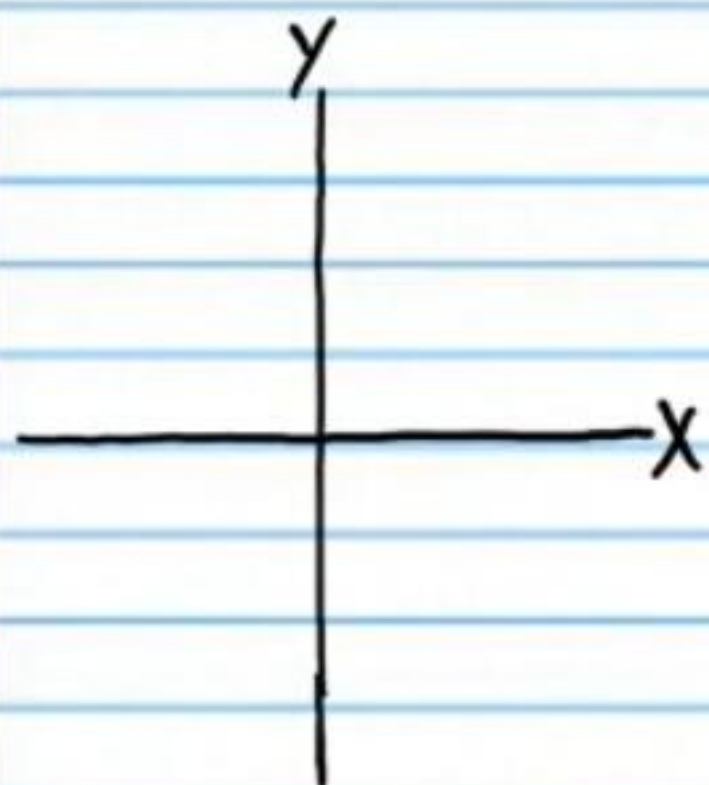


⑨ $(2, -3)$

Graph

Find r .

Find θ .

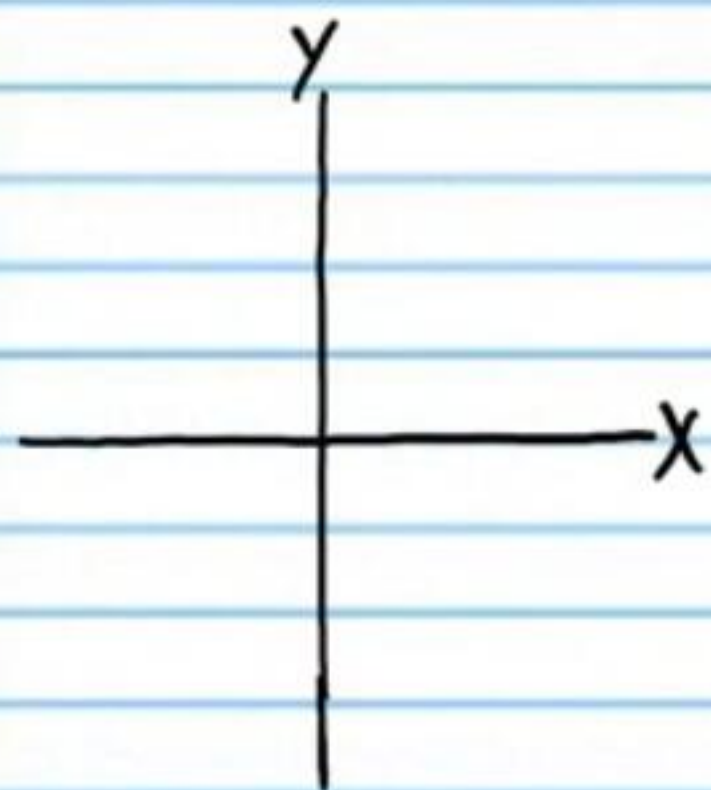


⑩ $(-1, 5)$

Graph

Find r .

Find θ .

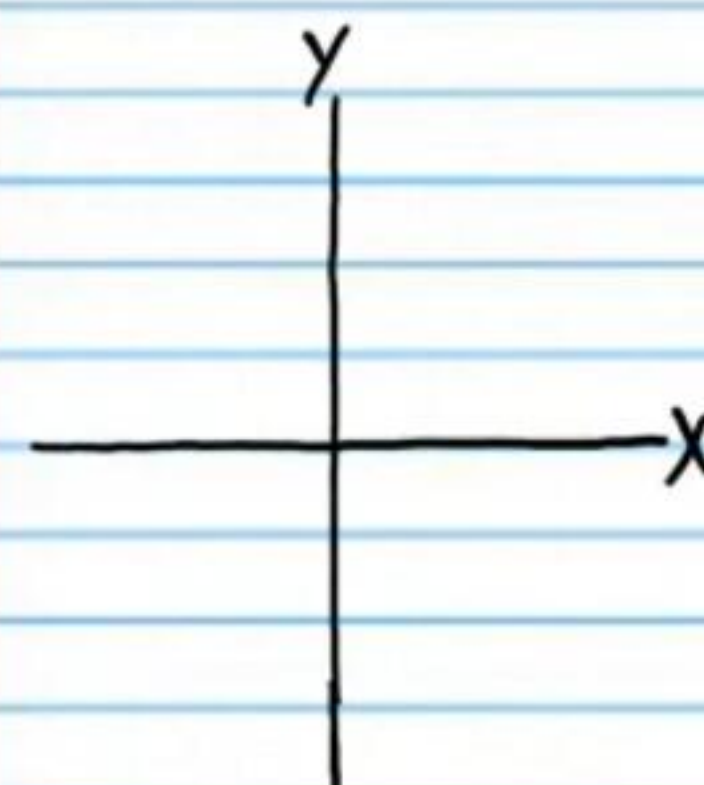


⑪ $(-7, -6)$

Graph

Find r .

Find θ .



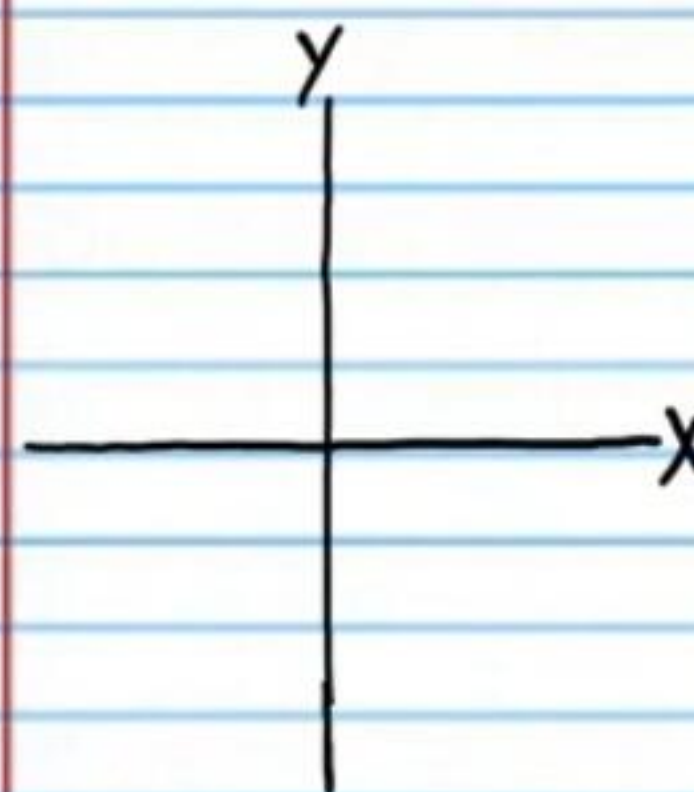
Do not use a calculator for the problems below.

• ⑫ $(3\sqrt{3}, 3)$

Graph

Find r .

Find θ .

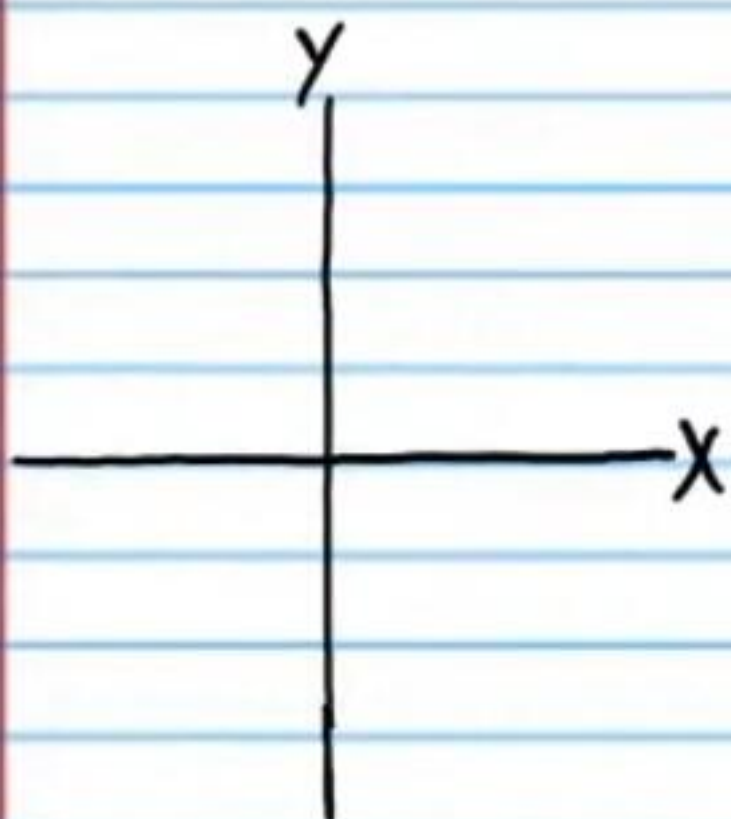


⑬ $(-1, -1)$

Graph

Find r .

Find θ .

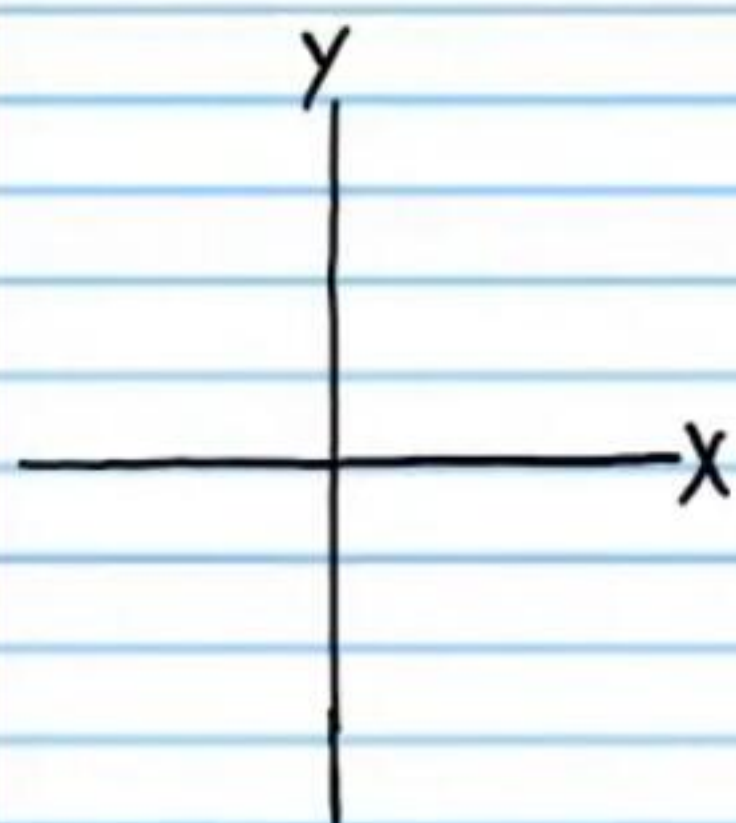


⑭ $(4, -4\sqrt{3})$

Graph

Find r .

Find θ .

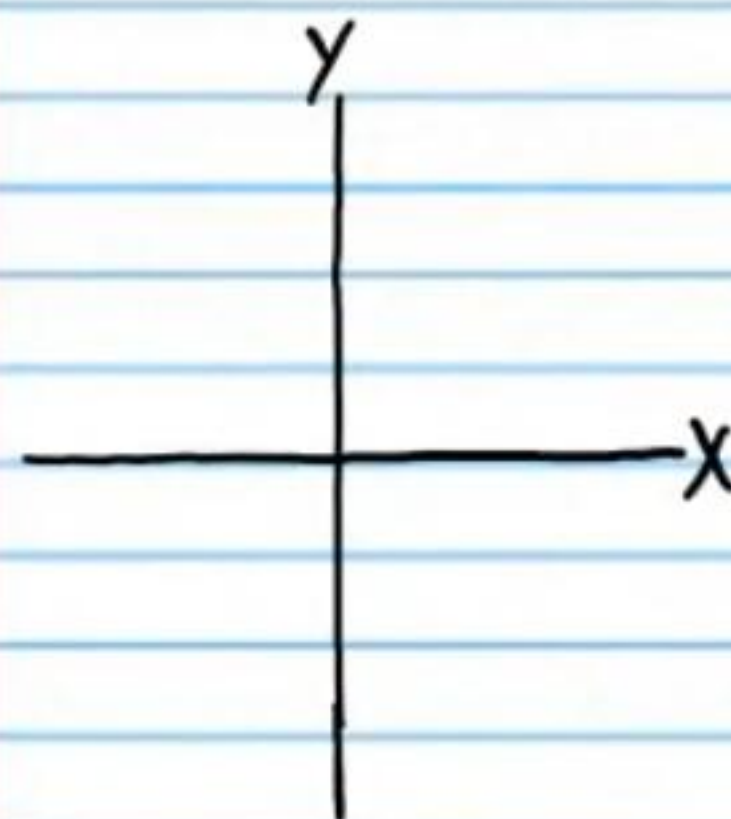


⑮ $(-10, 10)$

Graph

Find r .

Find θ .

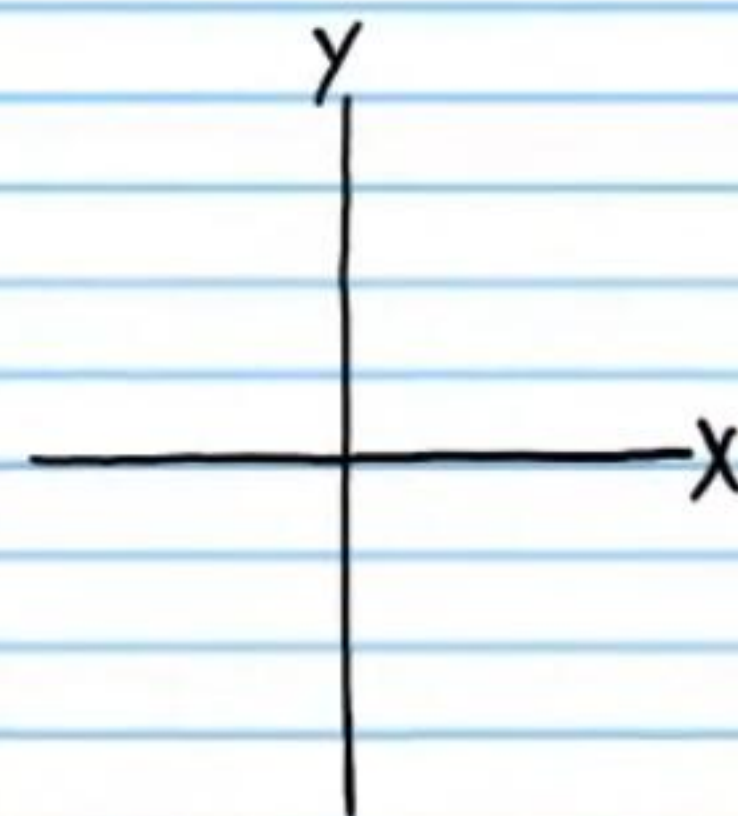


⑯ $(-4, 0)$

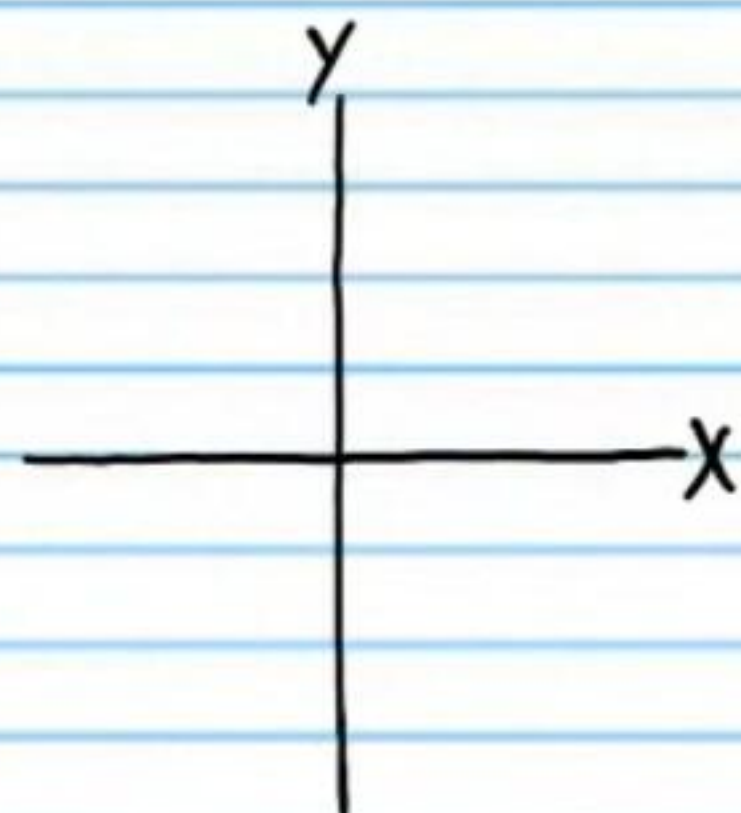
Graph

Find r .

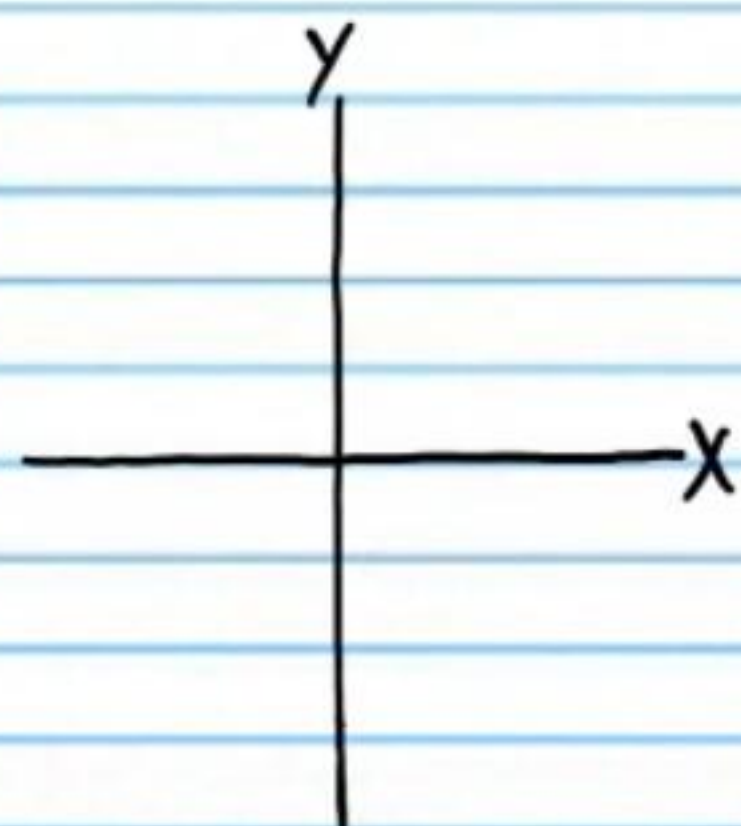
Find θ .



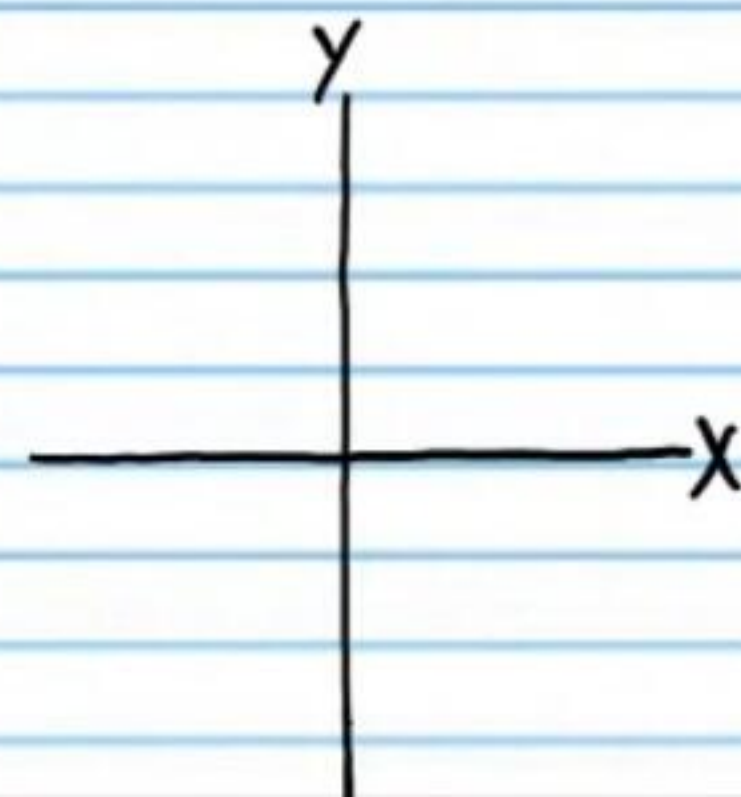
⑪ $(0, 2)$ Graph Find r . Find θ .



⑫ $(4, 0)$ Graph Find r . Find θ .



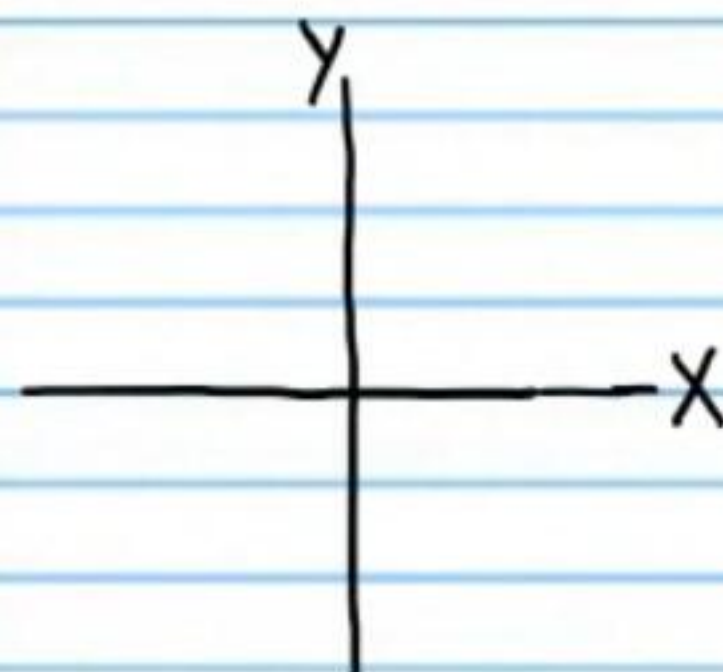
⑬ $(0, -3)$ Graph Find r . Find θ .



Convert the following polar coordinates to rectangular coordinates.
Round to the nearest hundredth.

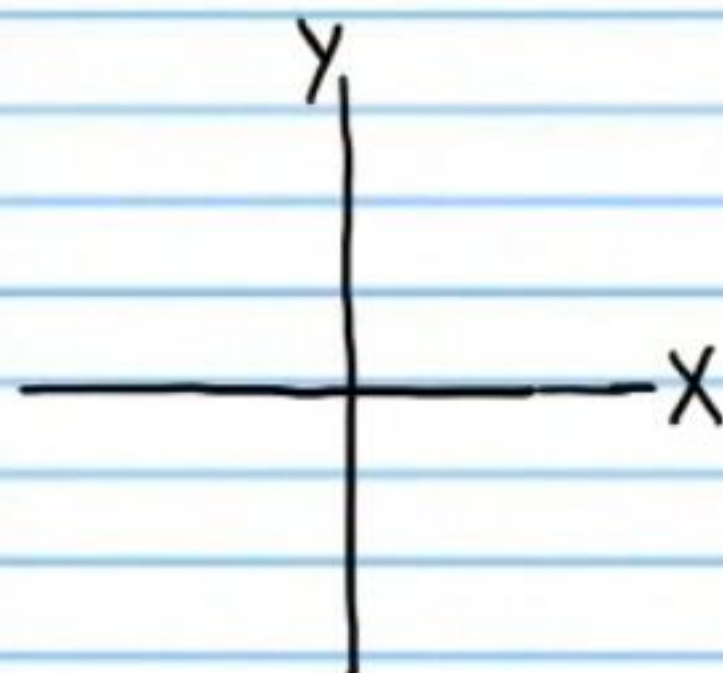
⑭ $(2, 35^\circ)$ Find x . Find y .

Graph



⑮ $(-3, -100^\circ)$ Find x . Find y .

Graph

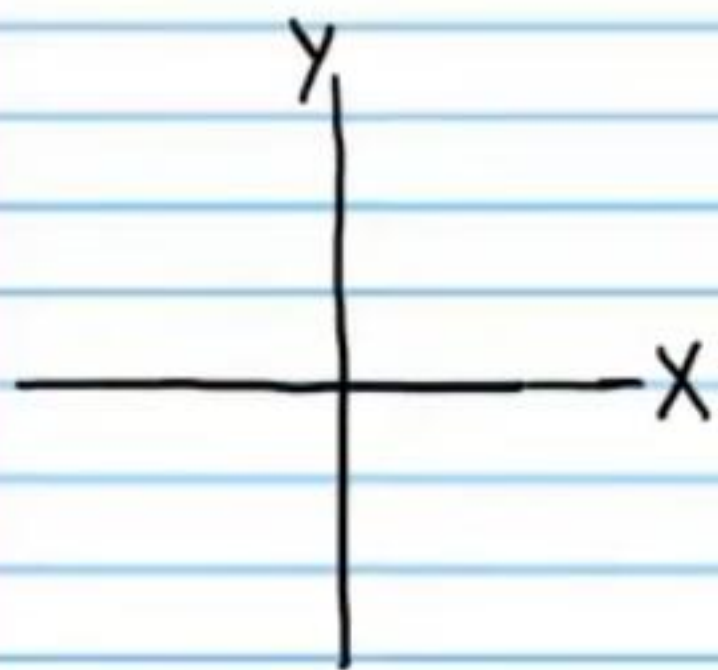


②② $(1, 129^\circ)$

Find x .

Find y .

Graph

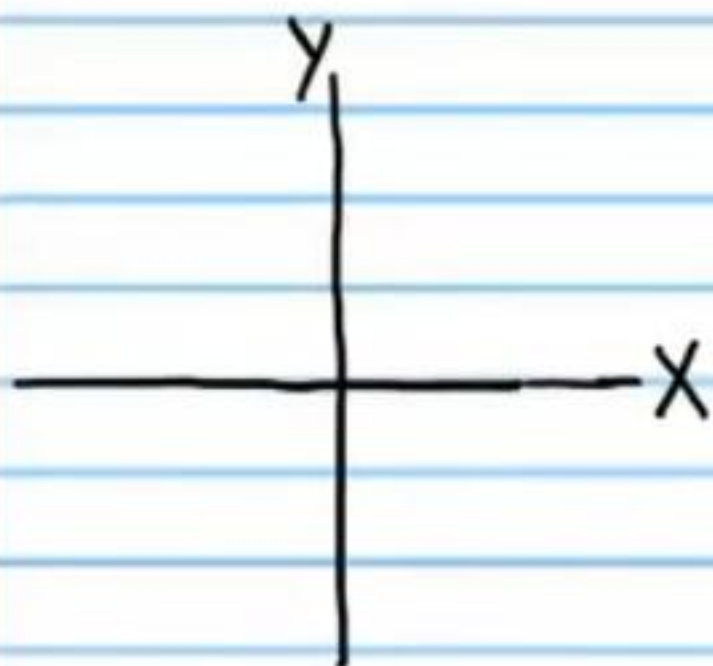


②③ $(-\frac{1}{2}, 147^\circ)$

Find x .

Find y .

Graph



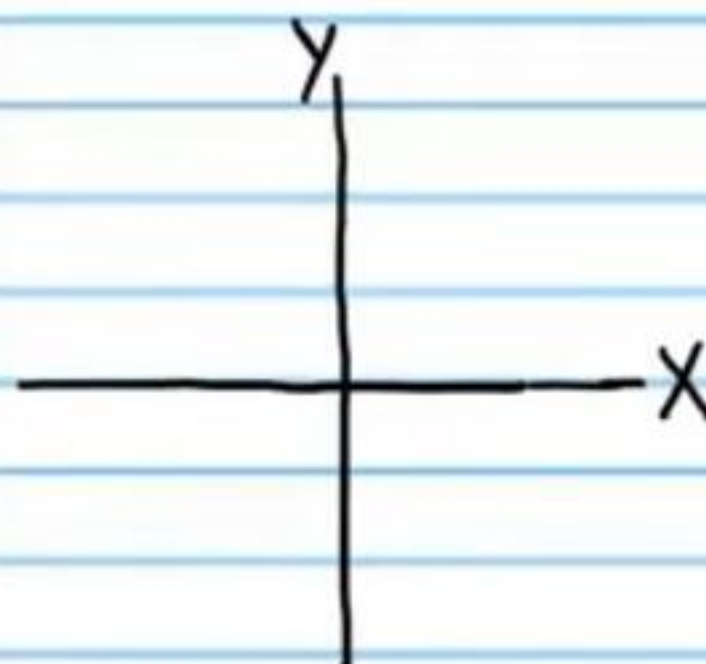
Do not use a calculator for the problems below.

②④ $(-3, 330^\circ)$

Find x .

Find y .

Graph

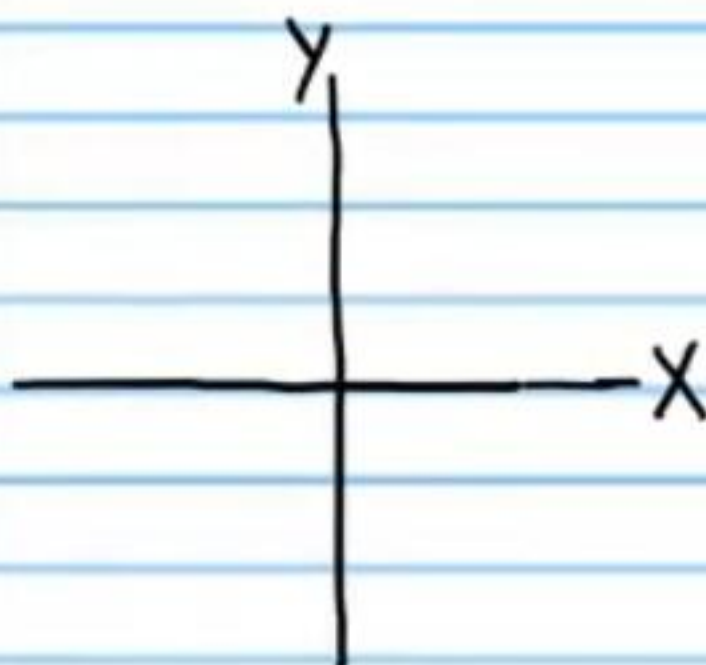


②⑤ $(-4\sqrt{2}, -225^\circ)$

Find x .

Find y .

Graph

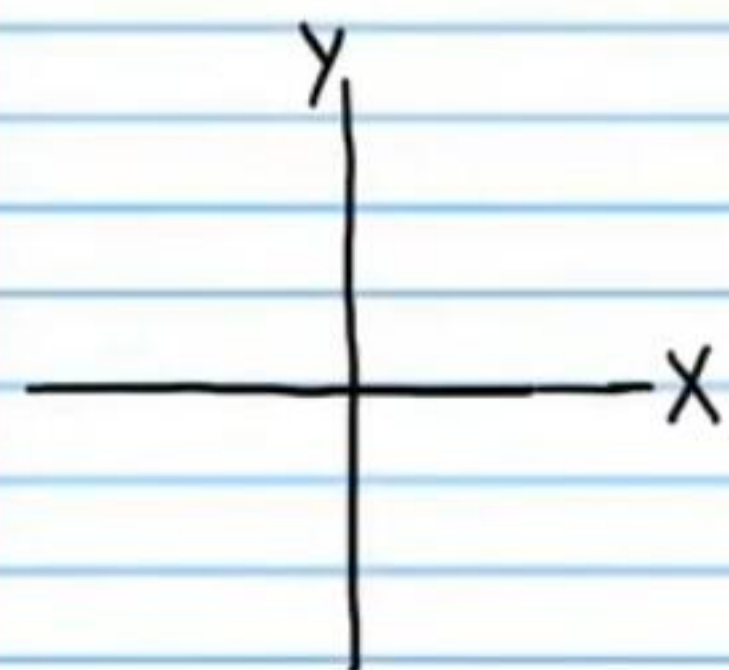


②⑥ $(1, 60^\circ)$

Find x .

Find y .

Graph

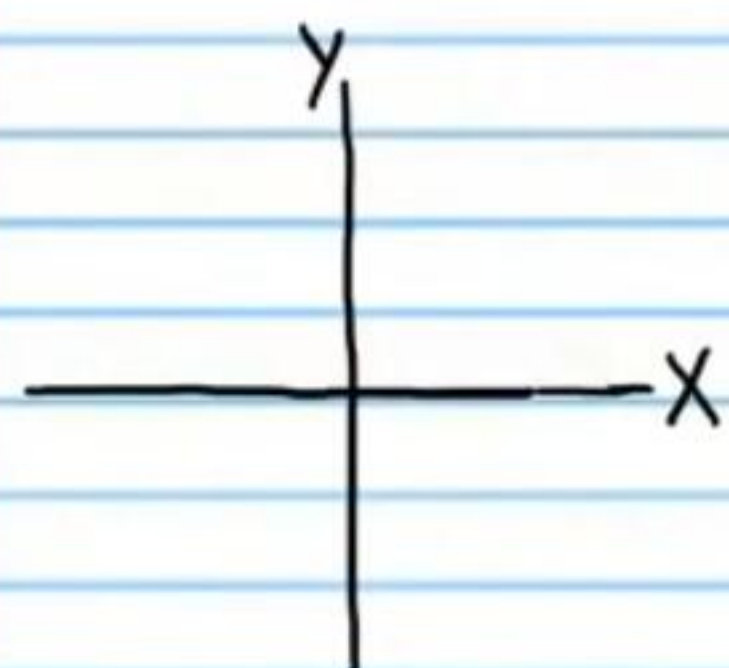


②⑦ $(\sqrt{2}, -135^\circ)$

Find x .

Find y .

Graph

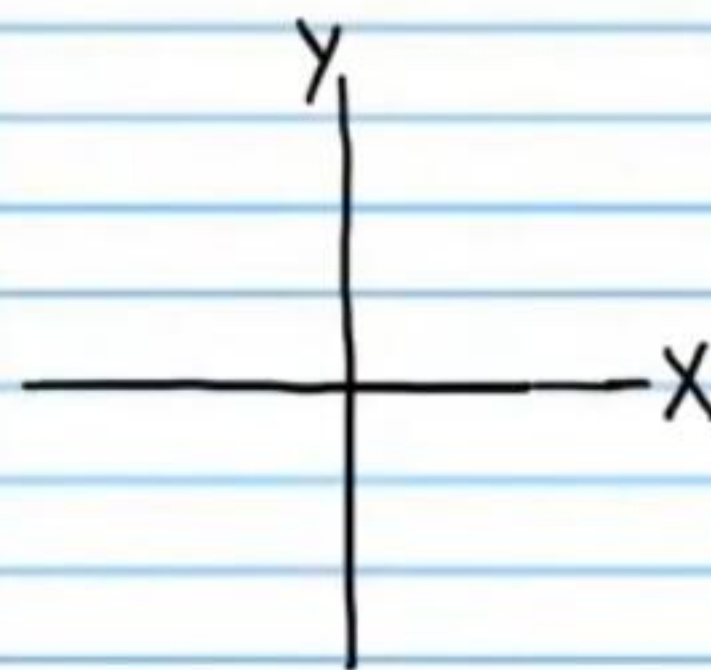


②⑧ $(2, 180^\circ)$

Find x .

Find y .

Graph

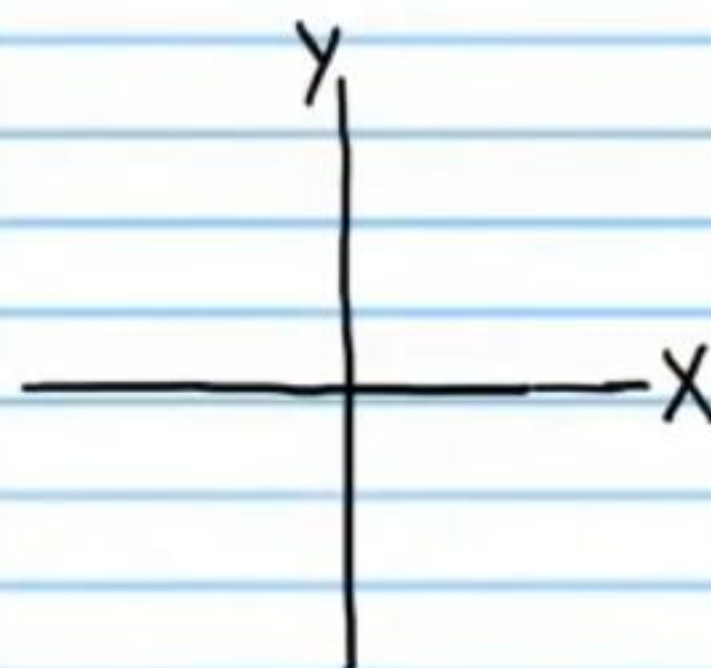


②⑨ $(1, 0^\circ)$

Find x .

Find y .

Graph

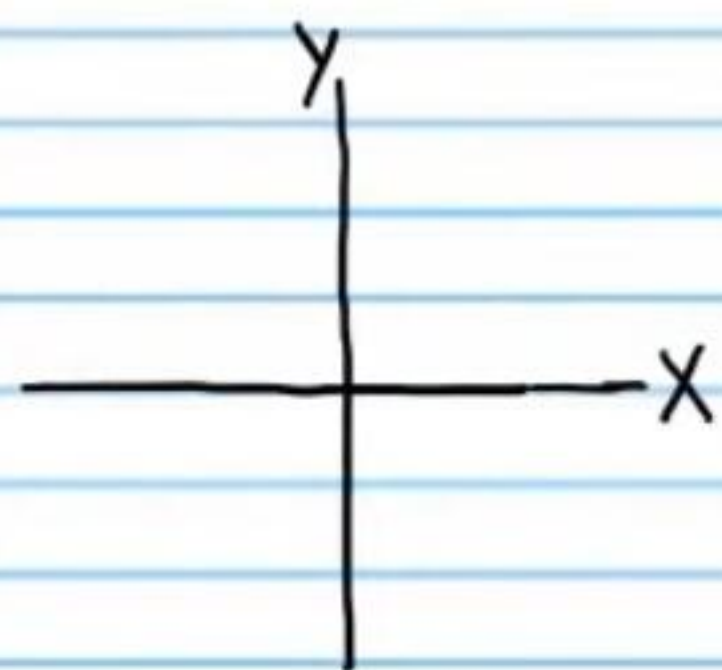


③① $(-5, 270^\circ)$

Find x .

Find y .

Graph

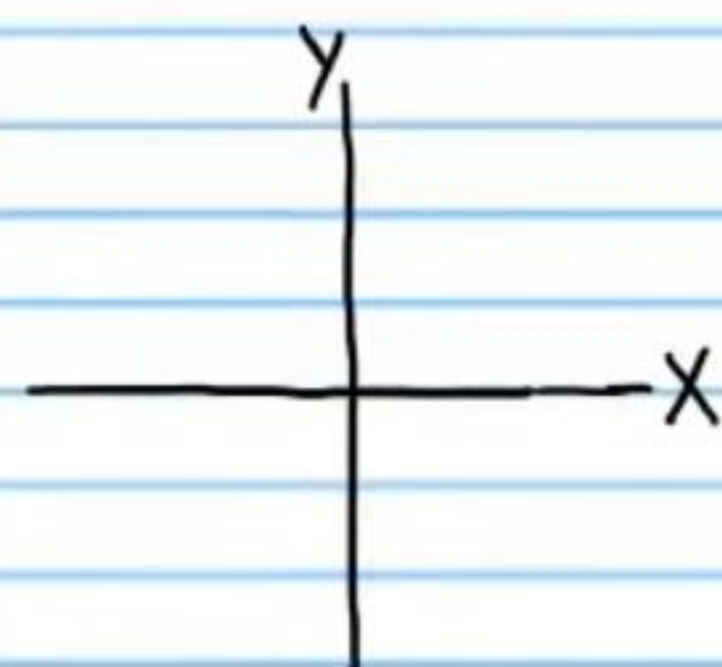


③② $(0, 205^\circ)$

Find x .

Find y .

Graph

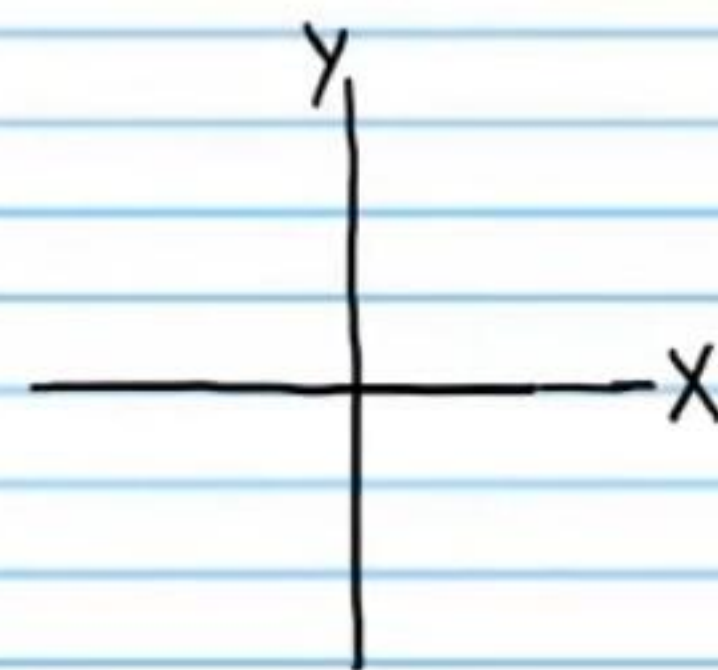


③③ $(8, 315^\circ)$

Find x .

Find y .

Graph

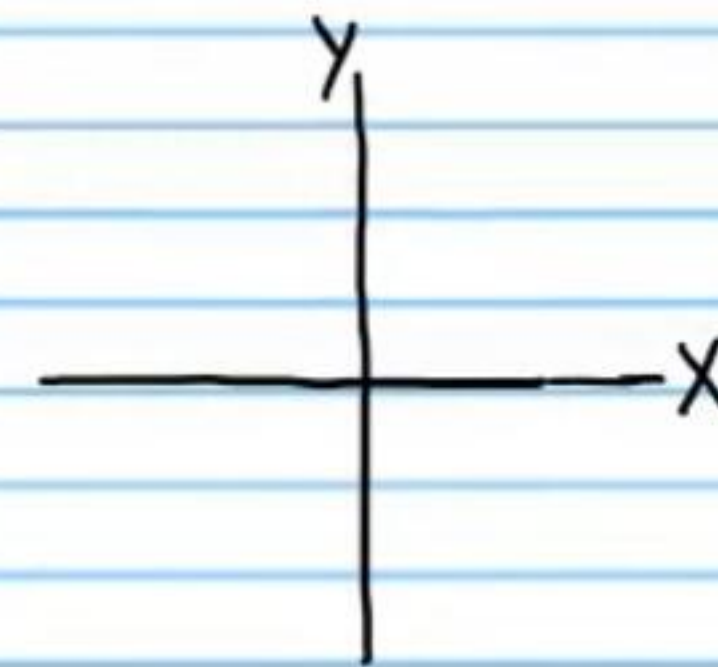


③④ $(10, 225^\circ)$

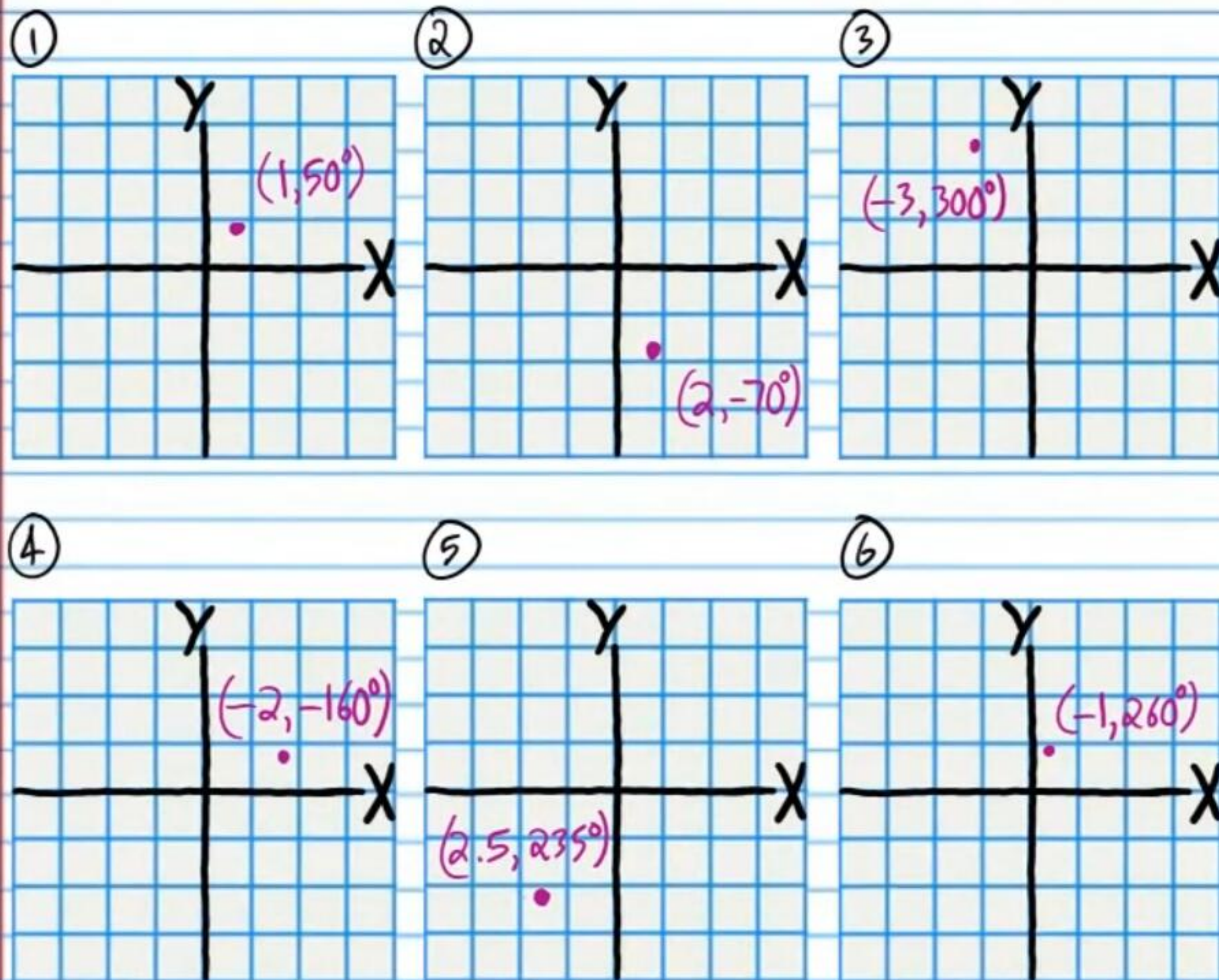
Find x .

Find y .

Graph



Answers



* Some solutions are shown below. Others are possible.

- ⑦ $(\sqrt{65}, -60.3^\circ)$ ⑩ $(\sqrt{26}, 101.3^\circ)$ or $(-\sqrt{26}, -78.7^\circ)$
- ⑧ $(\sqrt{61}, 39.8^\circ)$ ⑪ $(\sqrt{85}, 220.6^\circ)$ or $(-\sqrt{85}, 40.6^\circ)$
- ⑨ $(\sqrt{13}, -56.3^\circ)$ ⑫ $(6, 30^\circ)$

- ⑬ $(\sqrt{2}, 225^\circ)$ or $(-\sqrt{2}, 45^\circ)$ ⑮ $(10\sqrt{2}, 135^\circ)$ or $(-10\sqrt{2}, -45^\circ)$
- ⑭ $(8, -60^\circ)$ or $(8, 300^\circ)$ or $(-8, 120^\circ)$ ⑯ $(-4, 0^\circ)$ or $(4, 180^\circ)$
- ⑰ $(2, 90^\circ)$ or $(-2, -90^\circ)$ ⑱ $(4, 0^\circ)$
- ⑲ $(3, -90^\circ)$ or $(-3, 90^\circ)$ ⑳ $(1.64, 1.15)$
- ㉑ $(0.52, 2.95)$ ㉒ $(-0.63, 0.78)$
- ㉓ $(0.42, -0.27)$ ㉔ $(-\frac{3}{2}\sqrt{3}, \frac{3}{2})$
- ㉕ $(4, -4)$ ㉖ $(\frac{1}{2}, \frac{\sqrt{3}}{2})$
- ㉗ $(-1, -1)$ ㉘ $(-2, 0)$
- ㉙ $(1, 0)$ ㉚ $(0, 5)$
- ㉛ $(0, 0)$ ㉜ $(4\sqrt{2}, -4\sqrt{2})$
- ㉝ $(-5\sqrt{2}, -5\sqrt{2})$